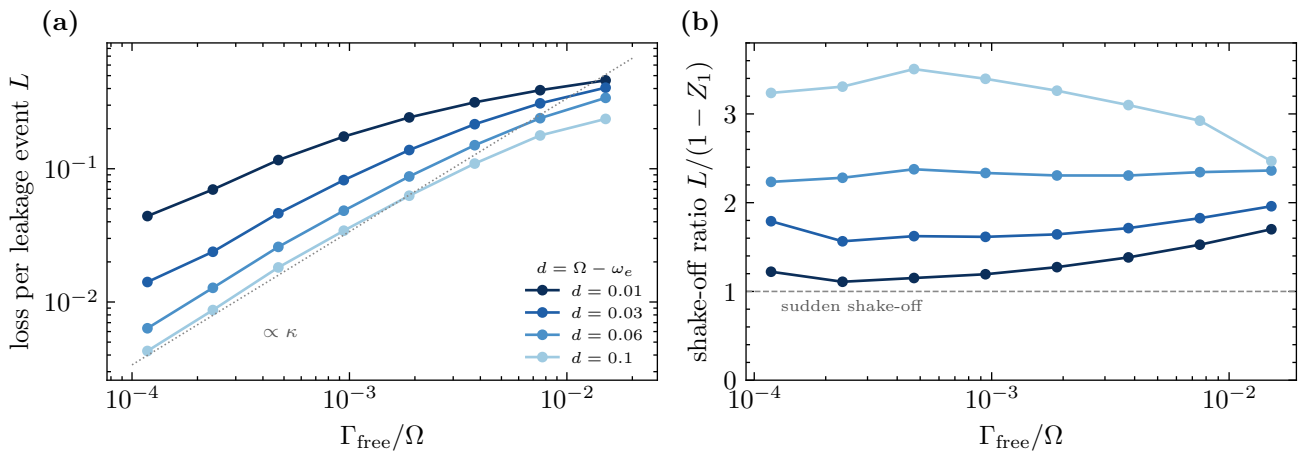

(a)
loss per leakage event L
10^{-1}
10^{-2}
$\propto \kappa$
$d = \Omega - \omega_e$
$d = 0.01$
$d = 0.03$
$d = 0.06$
$d = 0.1$
10^{-4}
10^{-3}
10^{-2}
$\Gamma_{\mathrm{free}}/\Omega$
(b)
shake-off ratio $L/(1 - Z_1)$
0
1
2
3
sudden shake-off
10^{-4}
10^{-3}
10^{-2}
$\Gamma_{\mathrm{free}}/\Omega$